Thermoelectric Properties of Co-doped CuAgSe

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Motivation

CuAgSe has been shown to exhibit a variety of unique low-temperature thermoelectric behaviors, includiing a strong thermoelectric figure of merit in samples doped with Ni. In order to further investigate CuAgSe's unusual thermoelectric properties and their dependence on chemical composition, we synthesized a series of CuAgSe samples doped with Co according to the following compositions: $Cu_{1-x}Co_xAgSe$ (x = 0.02, 0.05, 0.10) and measured their low temperature thermoelectric properties both in and out of a 9T magnetic field.

Sample Synthesis

Stoichiometric amounts of the constituent elements are placed in an alumina crucible that is sealed in an evacuated quartz ampule using a Hydrogen-Oxygen torch. The ampule is then loaded into a vertical furnace and melted according to a sequence obtained from [1]. The resulting melt is ball-milled and then the melting sequenece is repeated on the resulting powder.

Image 1: Quartz tube containing stoichiometric amounts of constituent elements in an alumina crucible being pumped out prior to sealing.

Image 2: Sample precursors inside sealed quartz ampule ready for melting.

Image 3: Polycrystalline sample obtained after melting in furnace.

Measurements

Polycrystalline samples were cut into bars using a diamond saw and placed onto the appropriate puck for measurement. All measurements were done in our Physical Property Measurement System (PPMS) by Quantum Design. Seebeck coefficient and thermal conductivity measurements were done using the PPMS Thermal Transport Option (TTO) and resistance measurements were done using a Linear Research (LR-700) AC resistance bridge. Ag epoxy was used to make contacts for TTO measurements and Ag paint was used for AC resistance measurements.

Image 4: Sample in TTO puck used to measure Seebeck and thermal conductivity.

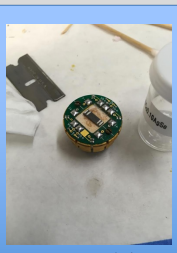

Image 5: Sample in AC resistance puck.

Image 6: Quantum Design PPMS used for all low-temperature measurements.

Acknowledgements

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Results

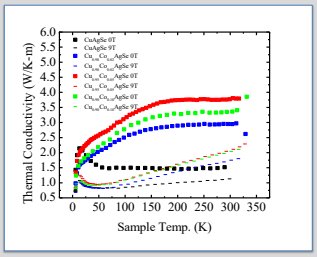

Figure 1: Thermal conductivity as a function of temperature both in and out of a 9T magnetic field.

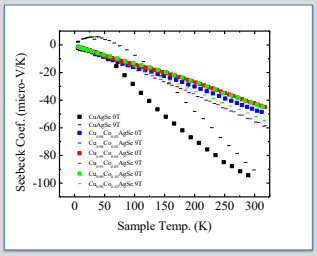

Figure 2: Seebeck coefficient as a function of temperature both in and out of a 9T magnetic field.

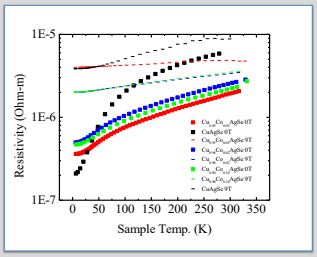

Figure 3: Electrical resistivity as a function of temperature both in and out of a 9T magnetic field.

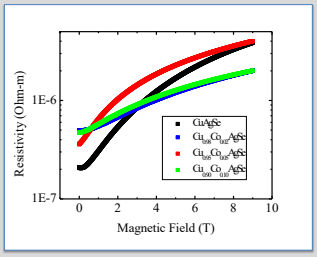

Figure 4: Resistivity as a function of magnetic field measured at 2K.

Discussion

- As Co concentration increases, resistivity decreases and a corresponding increase in thermal conductivity is observed. However, these changes begin to reverse in the 10% Co sample which may imply a decrease in mobility at high doping concentrations.
- The significant sensitivity of the Seebeck coefficient to an applied magnetic field is highly unusual and is consistent with the picture alluded to in [1] where the thermopower is strongly related to scattering phenomena.
- A significant enhancement in magnetoresistance such as what was observed in the Ni doped sample in [1] is not observed in our Co doped samples. This may imply that the Co ions are not acting as local moments in the same way the Ni ions appeared to in the earlier study.
- Next steps include:
  - Hall measurements to measure carrier density and mobility.
  - Heat capacity measurements to further investigate the thermodynamics of this fascinating system.
  - Doping with other 3d transition metals such as Fe or Cr.

References

1. Ishiwata, S. *et al. Nature Mater.* **12**, 512-517 (2013).
2. Wang, X. *et al. J. Mater. Chem. A*, 2015, **3**, 13662.
3. Hong, A. J. *et al. Solid State Ionics* **261**, 21-25 (2014).